

**GE MAGNET RIGGING**

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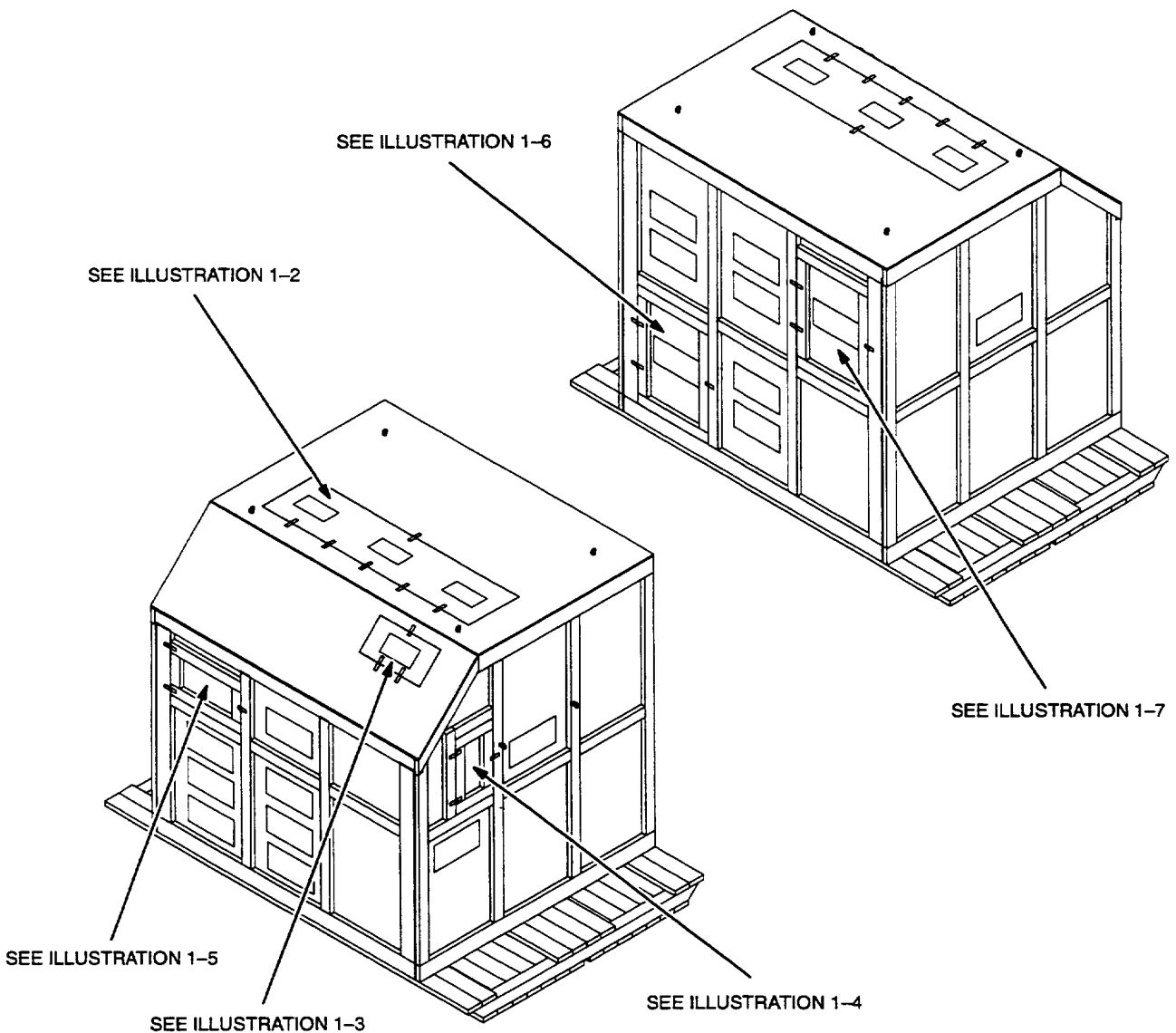
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## SECTION 1- SHIPPING / DELIVERY INSTRUCTIONS

### 1-1 INTRANSIT SERVICE

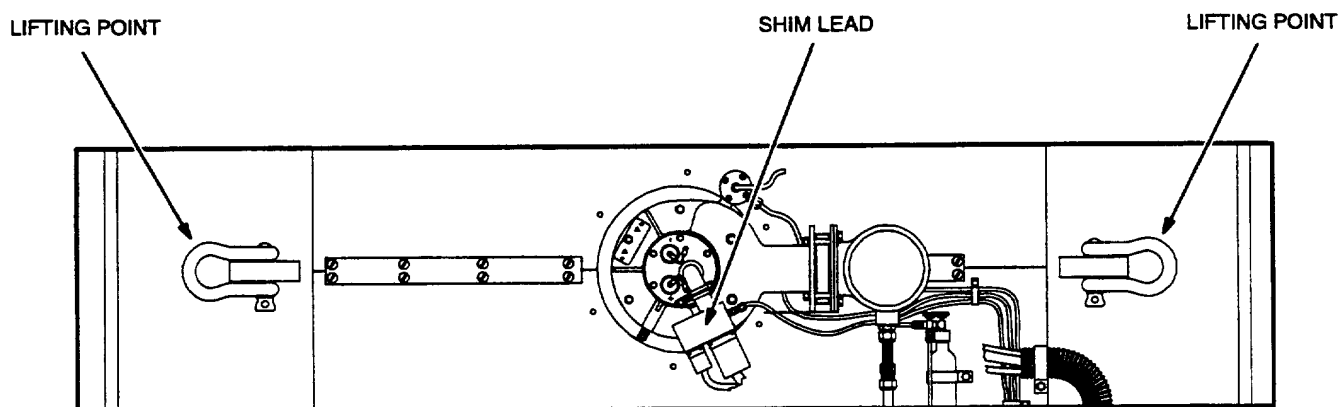
#### Note

The following information outlines provisions for filling and electrical / temperature checks while magnet is still in shipping crate. This will enable required service to be performed in transit based upon magnet shipping date. The following illustrations ( 1-1 through 1-7 ) show the magnet configuration behind each shipping crate access door. Access doors and hinges not shown for clarity.

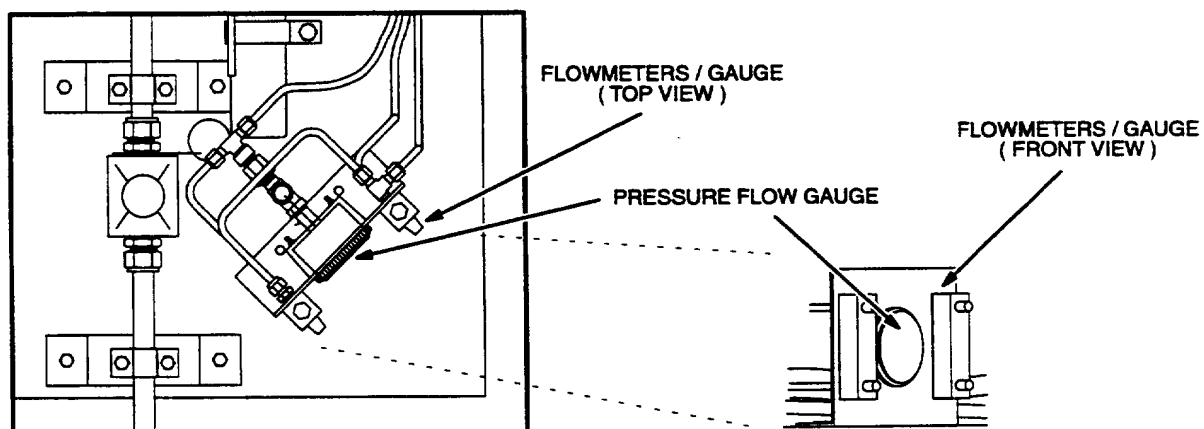


ACCESS DOOR LABELING ON CRATE  
ILLUSTRATION 1-1

## 1-1 INTRANSIT SERVICE (continued)

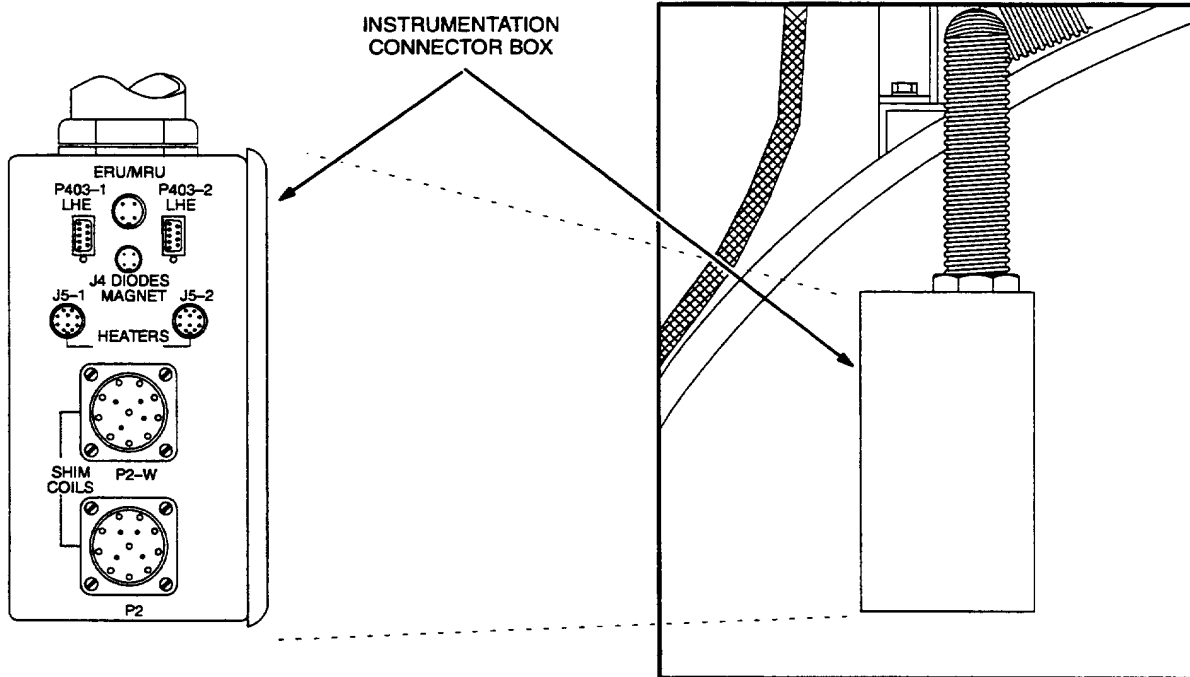


TOP ACCESS DOOR LABELED "LIFTING POINT" AND "SHIM LEAD ACCESS"  
ILLUSTRATION 1-2

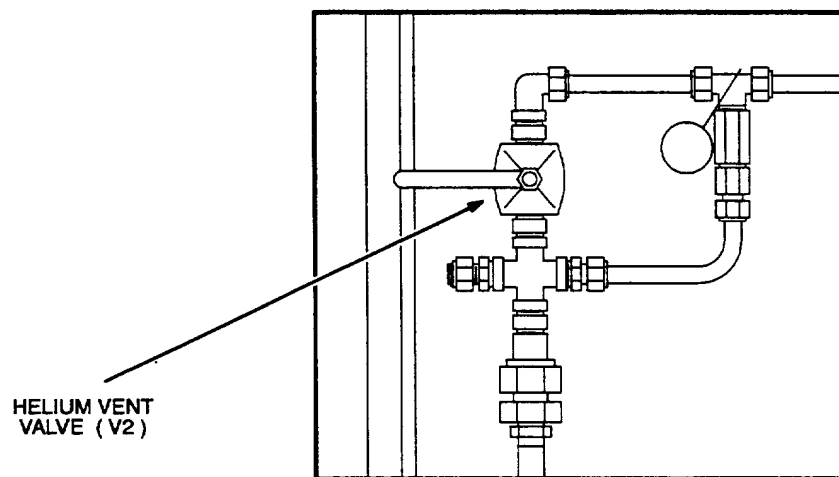


TOP ACCESS DOOR LABELED "PRESSURE FLOW GAUGE ACCESS"  
ILLUSTRATION 1-3

1-1 INTRANSIT SERVICE (continued)

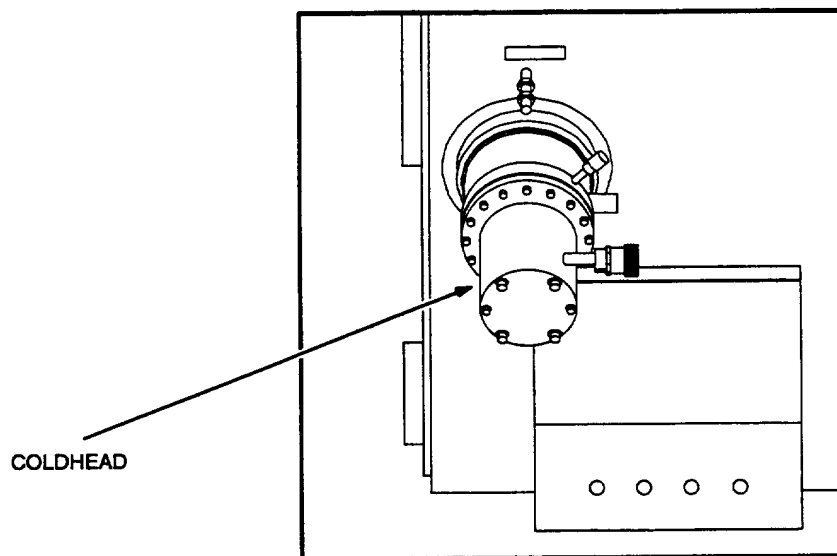


FRONT ACCESS DOOR LABELED "INSTRUMENT CONNECTOR"  
ILLUSTRATION 1-4

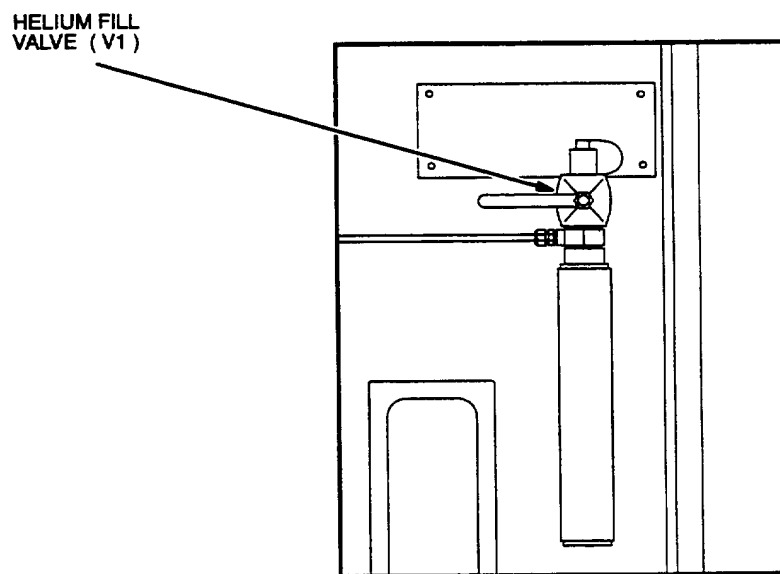


RIGHT SIDE ACCESS DOOR LABELED "V2 VENT PORT"  
ILLUSTRATION 1-5

1-1 INTRANSIT SERVICE (continued)



LEFT SIDE ACCESS DOOR LABELED "COLD HEAD ACCESS"  
ILLUSTRATION 1-6



LEFT SIDE ACCESS DOOR LABELED "V1 HELIUM FILL ACCESS"  
ILLUSTRATION 1-7

1-2 SHIPPING / HANDLING



Shipping and handling guidelines are provided in Table 1.

These guidelines must be followed to prevent any potential damage to the magnet during shipping and handling.

Review guidelines with Shipper / Riggers prior to transporting magnet.

TABLE 1  
MAGNET SHIPPING AND HANDLING INFORMATION

| MAGNET           | * MAXIMUM WEIGHT | ** MAXIMUM TILT | (1) ALLOWABLE SHIPPING MODES | (2) FORKLIFT CAPABILITY |
|------------------|------------------|-----------------|------------------------------|-------------------------|
| MAX / 0.5T Signa | 28,500 lbs.      | 30 deg.         | A, T, Tr, B                  | Yes                     |

| MAGNET           | SHIPPING CAPABILITY | (3) MAXIMUM TRANSIT TIME | MAXIMUM SHOCK LOADS | COMMENTS  |
|------------------|---------------------|--------------------------|---------------------|-----------|
| MAX / 0.5T Signa | Cold                | 21 days                  | 2 G's               | See Notes |

\* Includes weight of Shipping Crate ( 2,200 lbs. )

\*\* Tilt allowed when suspended by Lifting Shackles.

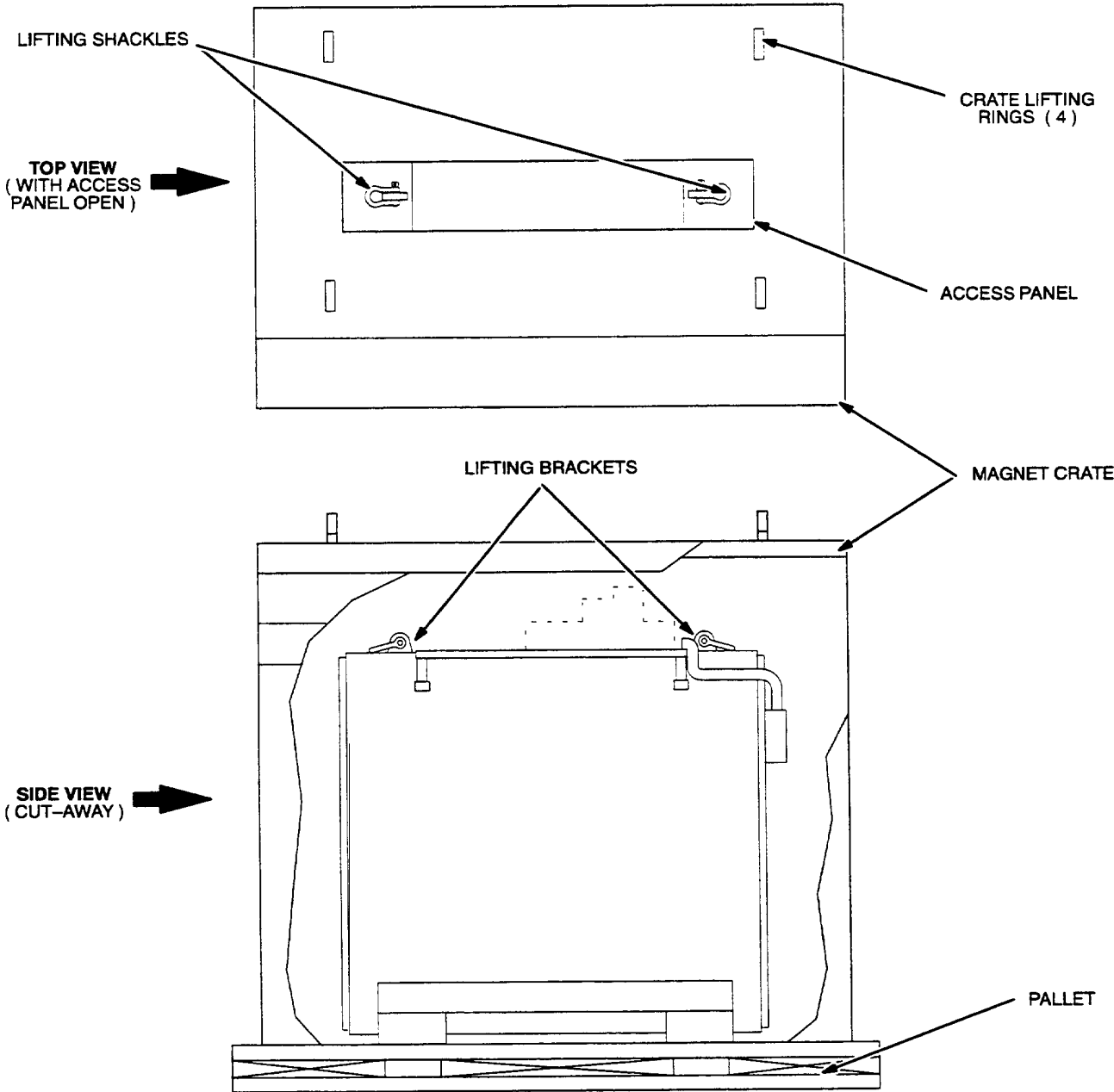
Notes:

- Key for Shipping mode symbols:
  - "A" Airplane (including any plane that has fuselage openings large enough to accept a magnet)
  - "T" Air ride Trailer (Any magnet transported on a non-air ride trailer must be identified and never used in a Mobile trailer.)
  - "Tr" Train
  - "B" Boat or ocean going ship
- Extreme care must be exercised during forklift operations. The magnet pallet/crate must be picked up from the sides only. The forks MUST be placed directly under the four ( 4 ) feet of the magnet. The magnet feet can be identified by the steel plates attached to the pallet.
- The elapsed time begins when the magnet leaves the Florence loading dock.

1-3 UNLOADING / LOADING MAGNET AND CRATE

Note

Lifting Shackles and Brackets are provided for magnet unloading / loading. See illustration 1-8.  
Shackle / Bracket access panel is provided, as shown in Illustration 1-8, for crated magnets.

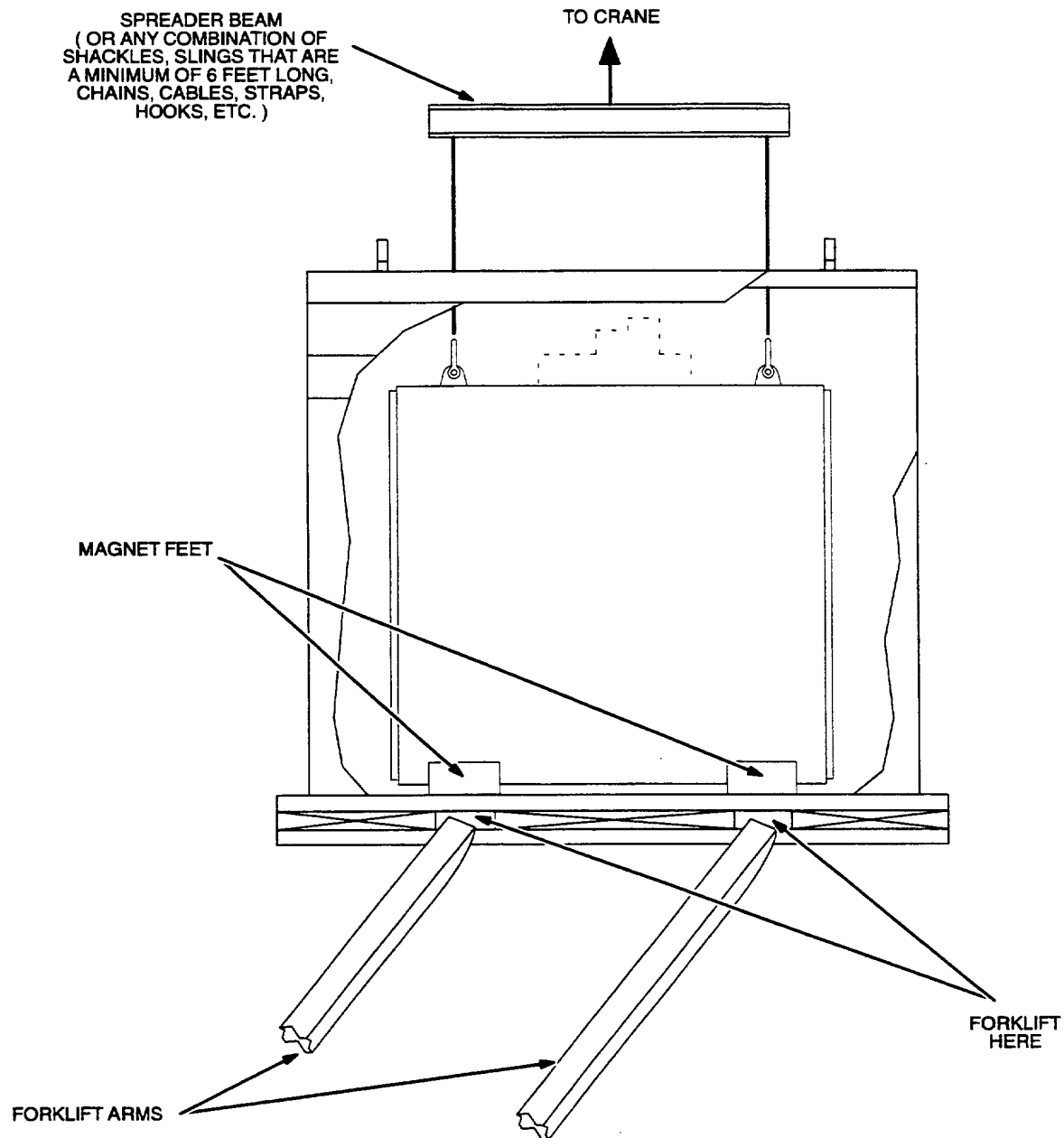


ACCESS PANEL, SHACKLE AND BRACKET LOCATIONS  
ILLUSTRATION 1-8

## 1-3 UNLOADING / LOADING MAGNET AND CRATE (continued)

**CAUTION**

Extreme care must be exercised during forklift operations. The magnet crate must be picked up from the sides only. The forks **MUST** be placed directly under the four (4) feet of the magnet. The magnet feet can be identified by the steel plates attached to the pallet. Do not lift crate with straps fed through the pallet. This will crush the crate.



FORKLIFT / CRANE LIFTING OPERATIONS  
ILLUSTRATION 1-9



**1-3 UNLOADING / LOADING MAGNET AND CRATE (continued)**

1. Position the Crane Hook centrally over the crated magnet to ensure a vertical lift force on the Lifting Brackets.



**It is important to lift the magnet smoothly to avoid impact or jolts to the system which may damage the magnet.**

2. Attach the rigging to the lifting shackles at each end of the crated magnet.

**Note**

Any combination of spreader beam, shackles and/or slings, 6 foot ( 1829 mm) minimum length, which can support a minimum of 28,500 pounds ( 12,928 Kg ) may be used.

3. Lift crated magnet with crane to clear trailer.
4. Lower the crated magnet onto a flat surface.

**1-4 PRE-DELIVERY INSTRUCTIONS**

**Make sure ALL site requirements / conditions, identified for the magnet in the site planning manual, are met before scheduling magnet delivery.**

**This will prevent installation delays, cryogen loss and resultant ongoing magnet quenches, potential damage, environmentally related problems and increased costs.**



**MAKE SURE BOTH MAGNET AND ROOM VENTING SYSTEMS ARE INSTALLED IN THE MAGNET ROOM, IN CONFORMANCE WITH THE SITE PLANNING MANUAL (REV. 4 AND ABOVE), PRIOR TO BRINGING THE MAGNET INTO THE ROOM. LARGE QUANTITIES OF GASEOUS HELIUM ARE DISCHARGED FROM THE MAGNET DURING INSTALLATION AND COMMISSIONING ACTIVITIES, WHICH WILL REQUIRE EXHAUSTING TO PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW CRYOGEN SAFETY MEASURES CONTAINED IN SECTION 5-3 OF THE INTRODUCTION ( CRYOGEN SAFETY ).**

1. Visit magnet site with rigging foreman before magnet delivery to plan the move.
2. Caution rigger that the magnet is extremely fragile. Sudden jolts can damage the magnet. Riggers aware of the cost of a magnet and its replacement usually use more care while handling magnet.
3. Ensure all roads and paths leading to exam room are level and free from obstacles and holes. Rigger will be required to construct platforms where needed.
4. If Roller Skids are to be used, have rigger bring eight steel plates, 36 in. x 12 in. x 0.25 in. (915 mm x 305 mm x 6.4 mm), to place along delivery route.
5. Mark the magnet location on the floor before delivery. Refer to architectural drawings.